

US Commodity Sources

A Breakdown of the USA's
Dependency on Competitors

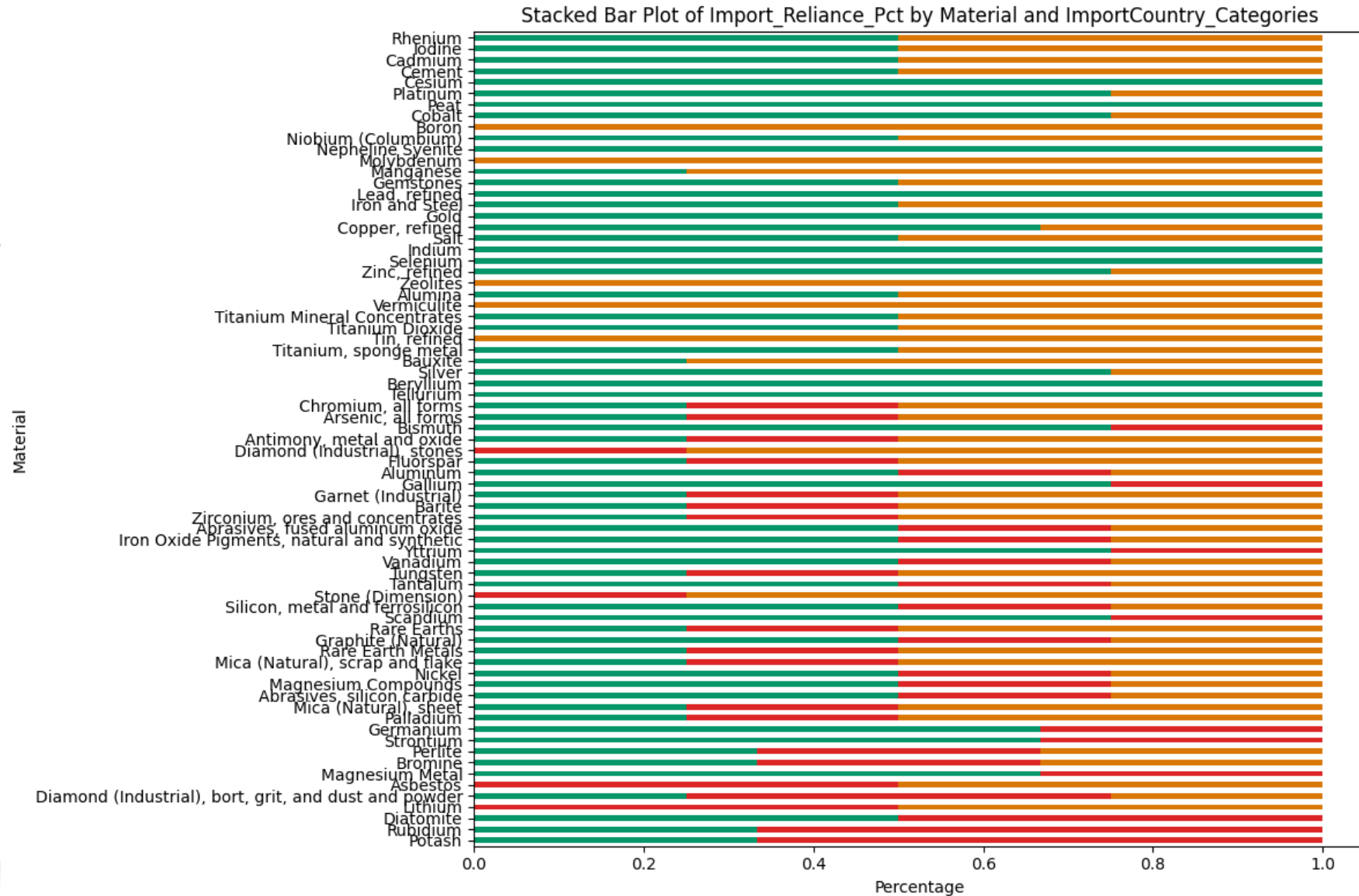
Shaya Engelman

Data 608 Story 7

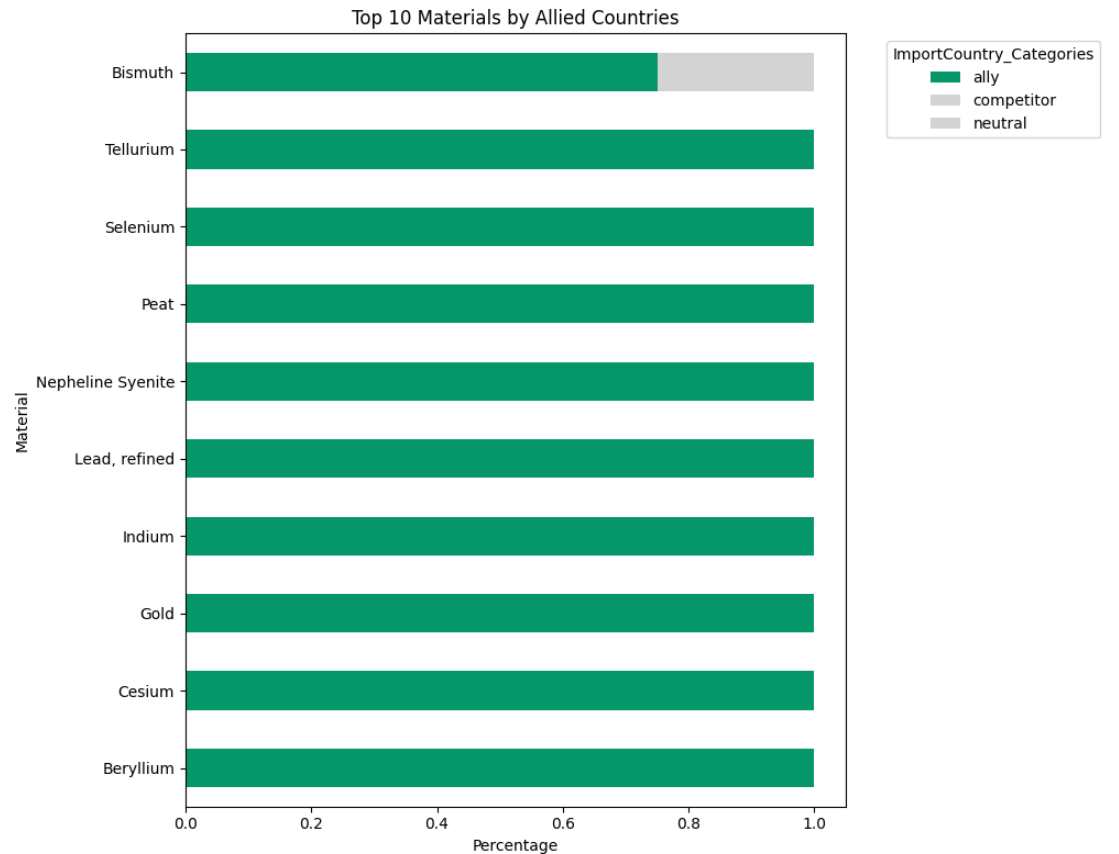
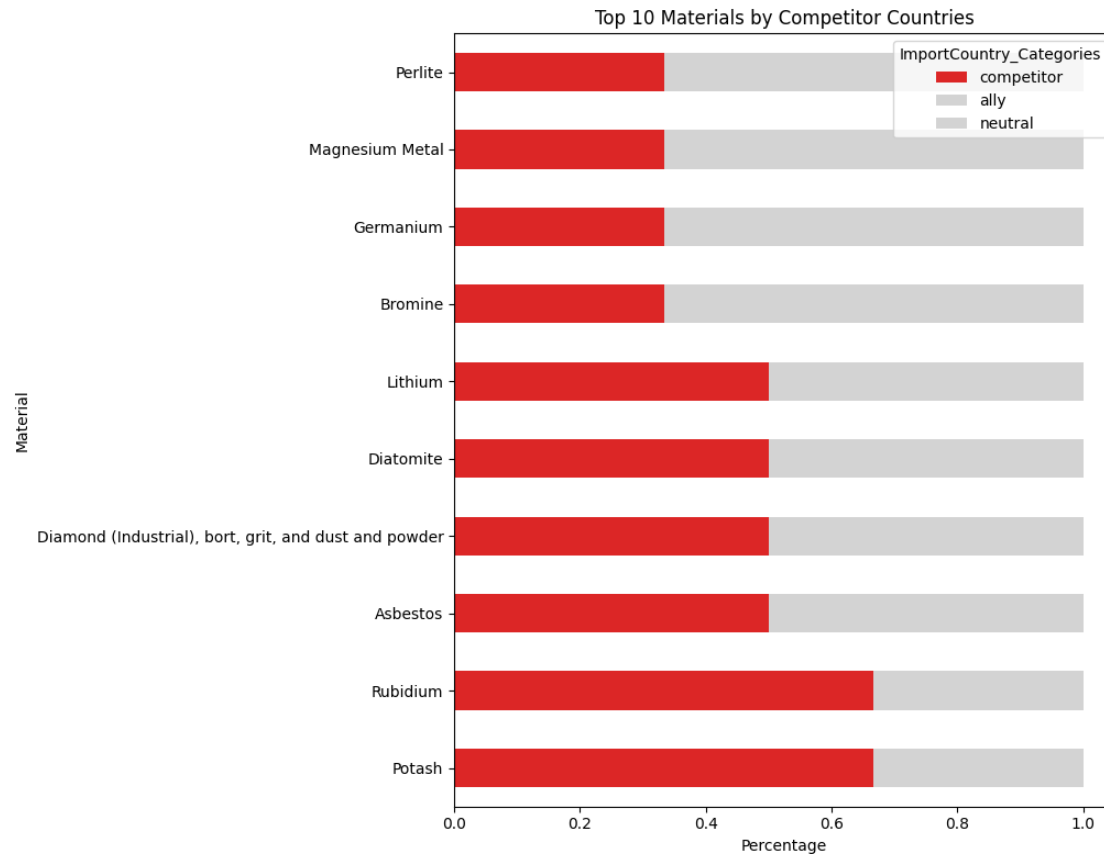


Stacked Barplots for each Commodity

- The US relies on allies for the majority of their trading needs.
- When not sufficiently able to source from allies, neutral countries are often used
- There do remain a few specific commodities that the vast majority or even all of it is sourced from competitors.
- This creates a dependency and major concern in times of strife

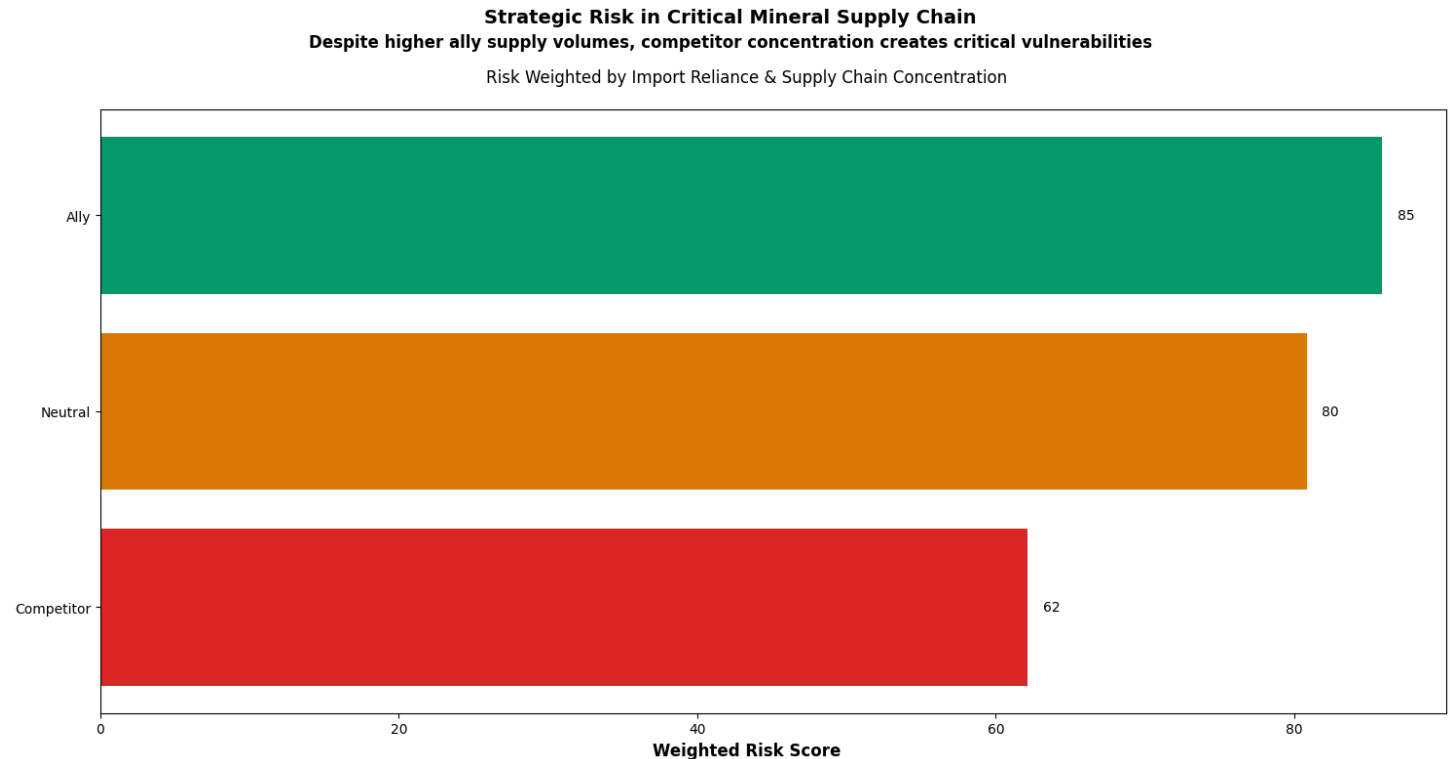


Top Safest and most Dependent Commodities



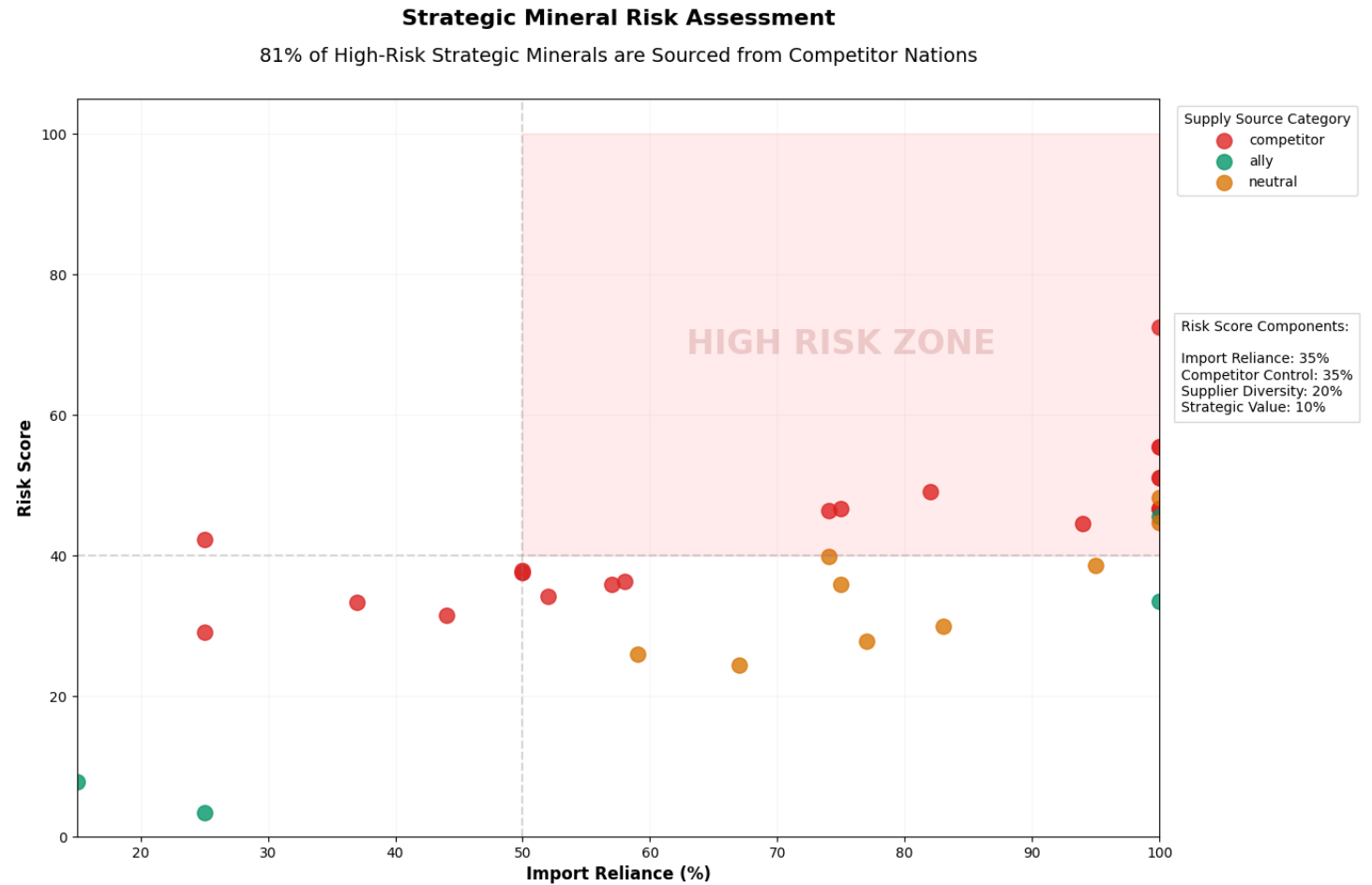
Overall Distributions by Weighted Risk of Critical Minerals

- **Weighting Factors:** The risk score is a composite measure combining four key factors:
- **Import Reliance (35%):** The percentage of a material's total imports from other countries.
- **Competitor Control (35%):** The degree to which competitors dominate the supply, adjusted by considering allies' mitigating effect.
- **Supplier Diversity (20%):** A measure of how diverse the supplier base is, penalizing reliance on single sources.
- **Strategic Value (10%):** An adjustment based on the strategic importance of the primary supplier, giving more weight to competitors and less to allies.
- Subset for the most critical minerals.
- As noted earlier, the US does get more from its allies and neutral countries than from its competitors.



Critical Minerals Broken Down Individually

- When subsetting for the most critical minerals- as defined by usgs.gov- there are some concerns
- 81% of the high risk critical minerals are sourced from competitor nations.
- Using the weighting used earlier, we see a few mineral in the high risk zone of having a very high import reliance percentage and a high risk score.



Sources

- Both the actual list of minerals and commodities and the list defining critical ones are from the following dashboard:

https://tableau.usgs.gov/views/MCSDashboardWorkbook_2024-01-30/MCSDashboard?%3Aembed=y&%3AisGuestRedirectFromVizportal=y#7

- Allies, Neutral, Competitor designation is based on NATO or other major alliances and traditional US adversaries.